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The effect of borate and silicate structure on thermal conductivity in the molten Na_2O–B_2O_3–SiO_2 system

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ABSTRACT

Thermal conductivity of molten Na_2O–B_2O_3–SiO_2 system was measured at various temperatures and for different compositions using the hot-wire method. It was observed that thermal conductivity decreases at higher temperatures. The calculated one-dimensional Debye temperature implies that the change of the phonon mean free path dominantly affects the thermal conductivity in the present measurements. Using solid state ^{11}B and ^{29}Si MAS NMR, the structure of the oxide system was confirmed. A conflicting effect of the R ratio (Na_2O/B_2O_3) on thermal conductivity was observed. For R below 0.5, owing to the increase in 4-coordinated boron, thermal conductivity increases with increasing R ratio. However, for R above 0.5, thermal conductivity decreases with increasing R ratio as a result of the depolymerization of silicate networks and the increase in non-bridging oxygen (NBO). Finally, the effect of the K ratio (SiO_2/B_2O_3) on thermal conductivity was considered. A higher K ratio causes an increase in thermal conductivity as a result of the increase in the total amount of the silicate structure and a more effective mixing between silicon and 4-coordinated boron.

1. Introduction

Owing to its low thermal expansion and excellent chemical durability, sodium borosilicate glass is widely used in laboratory apparatuses and household cookware [1]. However, it is also well-known for its complicated structural change, named as the "boron oxide anomaly". The coordination number of boron initially changes from 3 to 4 at high concentrations of alkali oxides; further addition of alkali oxides results in the change of 4-coordinated boron into 3-coordinated boron with the formation of non-bridging oxygen (NBO) and depolymerization of the silicate network structure. Thus, a considerable amount of studies [2–14] have been carried out to understand the short- and intermediate-range order of the sodium borosilicate system by using nuclear magnetic resonance (NMR), Raman spectroscopy, and molecular dynamic (MD) simulations.

As a result of the complicated structural change, the physical properties of the sodium borosilicate system cannot be estimated by a simple linear approximation based on chemical compositions [15]. Several studies have been conducted to elucidate the relations between the structure of the sodium borosilicate system and the physical properties [16–21]. Shiraishi et al. [16] reported that the measured viscosity of glassy Na_2O–B_2O_3–SiO_2 systems initially increased with increasing B_2O_3 content until the Na_2O/B_2O_3 ratio reached the unity, and subsequently decreased. They attributed the dependence of the viscosity on the Na_2O/B_2O_3 ratio to the change of the boron coordination number. Such a conflicting effect of B_2O_3 on the viscosity of the Na_2O–B_2O_3–SiO_2 system was also reported by Zhang and Reddy [17], who measured the viscosity in the molten sodium borosilicate system. They observed that the addition of B_2O_3 resulted in an increase in viscosity at low SiO_2/Na_2O ratio, but in a decrease in viscosity at high SiO_2/Na_2O ratio. Ghoneim and Halawa [18] measured the thermal conductivity of the sodium borosilicate glass at room temperature and found that the formation of both the BO_4 and SiO_4 structure groups results in an increase in the thermal conductivity along with an extension of the phonon mean free path.

Even if other physical properties have been measured across various temperature ranges; from glass to melts, the thermal conductivity has been mainly studied in the glass state. Although understanding the thermal conductivity of the molten oxide is also important for the thermal processing of the glass [22], the thermal conductivity measurement in the molten oxide system has been limited because of the large convection and the radiation effects [23]. However, owing to the several modifications of the non-steady state method, the thermal conductivity has been successfully measured in the molten oxide system over the last few decades [24–33]. Similar to other physical properties of the molten oxide system, the thermal conductivity is closely related to the network structure (i.e., silicate, aluminate, and borate structure.)

Recently, Kim and Morita [25] measured the thermal conductivity in the molten Na_2O–SiO_2 and Na_2O–B_2O_3 system. In the Na_2O–SiO_2 binary system, a linear relationship between thermal conductivity and viscosity was obtained, indicating the dominant effect of the Q^4 unit, which is the SiO_4 tetrahedron with four bridging oxygen atoms and zero NBO

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atoms. In the $Na_2O–B_2O_3$ system, the positive relationship between thermal conductivity and the relative fraction of 4-coordinated boron was observed in the region where the tetraborate unit is dominant. However, the simultaneous effect on the thermal conductivity of both the relative fraction of 4-coordinated boron and silicate structures in the molten oxide system has not been reported yet.

Although Kim and Morita [26] measured thermal conductivity in the $CaO–SiO_2–B_2O_3$ mold flux system, it mainly consisted of the metaborate units formed by asymmetric 3-coordinated boron due to the high CaO concentration and the low B_2O_3 concentration. For this reason, the effect of 4-coordinated boron on thermal conductivity could not be appropriately evaluated along with silicate network structure.

Therefore, in the present work, the thermal conductivity of the molten $Na_2O–B_2O_3–SiO_2$ system was measured within the tetraborate or diborate dominant region in order to evaluate the effect of both relative fraction of 4-coordinate boron and silicate network structure. The thermal conductivity was measured by adopting a hot wire method. The structure of borate and silicate structure was evaluated using ^{11}B and ^{29}Si magic angle spinning (MAS) NMR.

2. Experimental procedures

2.1. Sample preparation

The experimental compositions of the present work are shown in Fig. 1. The samples were prepared with fixed K ratios (SiO_2/B_2O_3) of 0.5, 1.0, and 2.0, and various R ratios (Na_2O/B_2O_3). To investigate the glass structure of the present system, the experiments were performed outside the "No glass" region [34].

Reagent grade SiO_2, B_2O_3, and Na_2CO_3 were weighed and subsequently ground in an agate mortar for 20 min to obtain a homogeneously distributed mixture. The powder mixture was then transferred to a platinum crucible and kept in the temperature range of 1423–1573 K, depending on its melting temperature, to enable it to decarbonate, dehydrate, and homogenize. After 2 h, the melt was quenched on a water-cooled copper plate, and then it was finely crushed. The product was transferred to a Pt-10%Rh crucible (inner diameter (ID): 32 mm; outer diameter (OD): 38 mm; height: 70 mm), and the thermal conductivity was measured by the hot-wire method in the molten state.

After the measurement, the solidified sample was re-melted at 1373 K, quenched by pouring it onto a water-cooled copper plate, and then quickly covered with a copper block. The obtained glass was ground and sieved under 100 μm ϕ mesh. The final compositions of Na_2O and B_2O_3 were analyzed by inductively coupled plasma atomic

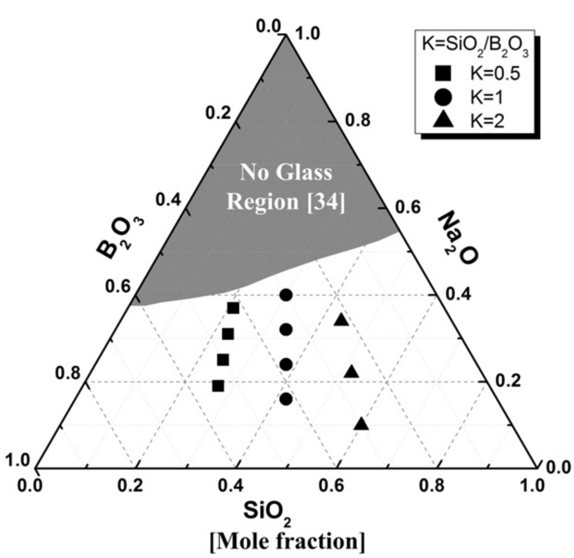

Fig. 1. Experimental compositions of the $Na_2O–B_2O_3–SiO_2$ system.

Table 1
Initial and final chemical compositions of the $Na_2O–B_2O_3–SiO_2$ systems used in the present work.

	Initial composition mol% (weighed)			R (Na_2O/B_2O_3)	K (SiO_2/B_2O_3)	Final composition mol% (analyzed)		
	Na_2O	SiO_2	B_2O_3			Na_2O	SiO_2	B_2O_3
1	37.0	21.0	42.0	0.88	0.5	35.2	19.8	45.0
2	31.0	23.0	46.0	0.67	0.5	30.6	23.0	46.4
3	25.0	25.0	50.0	0.50	0.5	24.3	23.2	52.5
4	19.0	27.0	54.0	0.35	0.5	18.1	25.4	56.5
5	40.0	30.0	30.0	1.33	1.0	37.4	30.6	32.0
6	32.0	34.0	34.0	0.94	1.0	29.6	34.9	35.5
7	24.0	38.0	38.0	0.63	1.0	23.4	37.9	38.7
8	16.0	42.0	42.0	0.38	1.0	15.2	42.3	42.5
9	34.0	44.0	22.0	1.55	2.0	32.8	44.5	22.7
10	22.0	52.0	26.0	0.85	2.0	21.1	52.2	26.7
11	10.0	60.0	30.0	0.33	2.0	9.4	59.8	30.8

emission spectroscopy (ICP-AES; SPS7700, SII NanoTechnology, Tokyo, Japan), and the SiO_2 content was analyzed using a gravimetric analysis technique. The weighted initial and analyzed final compositions are listed in Table 1.

For the ^{29}Si MAS NMR measurement, additional samples containing 0.1 mol% Fe_2O_3 to reduce the spin–lattice relaxation time were prepared [35]. The procedure followed to prepare the samples was the same as that described above.

2.2. Thermal conductivity measurement and error on the measurement

The Pt-10%Rh crucible, filled with approximately 90 g of the pre-melted sample, was placed in a vertical furnace equipped with a SiC heating element. The sample was held for 1 h at 1273 K to obtain a homogeneous phase, and then the thermal conductivity of the $Na_2O–B_2O_3–SiO_2$ system was measured from 1273 K to the liquidus temperature at intervals of 50 K. The temperature of sample was controlled by the PID (proportional integral differential) controller and calibrated B-type thermocouple (Pt-30%Rh/Pt-6%Rh) within ± 3 K. To ensure the thermal equilibrium of the system, the furnace temperature was reduced at a rate of 3 K/min and held for 15 min at the target temperature. At each temperature, the thermal conductivity was measured three times at intervals of 5 min to examine the reproducibility. The random error was determined by the standard deviation of the repetitive measurements [36], and it is displayed in the figures as error bar. During the thermal conductivity measurement, constant current, applied by galvanostat, was determined between the standard resistor of 0.1 Ω (Yokogawa 2792, Yokogawa, Tokyo, Japan); accuracy of 0.01%. Any voltage change between the potential leads at a four-terminal hot-wire sensor was recorded using digital multimeter (Keithley 2000, Keithley instruments, Cleveland, OH, USA); accuracy of 0.002%. It should be noted that the heat can be transferred by radiation from the Pt-13%Rh hot wire to the molten oxide, and contribution of radiation would be considerable due to the high temperature. Using the Stefan–Boltzmann law for gray-body radiation, the heat radiation at the surface of hot wire was estimated. During the calculation, emissivity of Pt-13%Rh wire was extrapolated on the basis of empirical equation of emissivity of Pt-10%Rh wire [37]. The calculation indicates that approximately 0.8% of total heat is transferred by radiation at 1273 K. In Table 2, systematic errors on present measurement are summarized.

Table 2
Estimation of the systematic errors on the thermal conductivity measurement.

Uncertainty component	Relative uncertainty
Temperature calibration	0.2%
Accuracy of standard resistor	0.01%
Resolution of digital multimeter	0.002%
Heat by radiation	0.8%

Table 3
Experimental conditions for the ^{11}B and ^{29}Si MAS NMR measurements.

Nuclear property	^{11}B	^{29}Si
Nuclear spin	3/2	1/2
Larmor frequency [MHz]	160.4	99.4
RF pulse intensity [kHz]	66	50
Repetition times	64	1000
Flip angle [rad]	π/6	π/2
Delay time [s]	5	20
Spinning rate [kHz]	16	16

The detailed experimental conditions and procedures of the hot-wire method have been described elsewhere [26,27].

2.3. Structural investigation

After obtaining confirmation of the non-crystalline state by X-ray Diffraction (XRD; RINT 2500, Rigaku, Tokyo, Japan), a structural investigation of the samples was conducted by using MAS NMR.

Solid state ^{11}B and ^{29}Si MAS NMR spectra were recorded at 11.74T using a Fourier transform (FT) NMR spectrometer (ECA-500, JEOL, Tokyo, Japan). The Larmor frequency was 160.4 MHz for ^{11}B MAS NMR and 99.4 MHz for ^{29}Si MAS NMR. Each sample was placed in a 4 mm ZrO_2 holder and spun at 16 kHz. Using a saturated H_3BO_3 solution (19.49 ppm) and 3-(Trimethylsilyl)propionic-2,2,3,3-d_4 acid sodium salt (98 atom % D) (1.445 ppm), the standards of ^{11}B and ^{29}Si MAS NMR were calibrated. The detailed conditions adopted for the MAS NMR measurements are reported in Table 3.

After the measurements, the ^{29}Si MAS NMR data were fitted by a Gaussian function with the 'OriginPro (Ver.8.5)' software to obtain structural information.

3. Results

3.1. Effect of the temperature on the thermal conductivity

Fig. 2 shows the temperature dependence of the thermal conductivity with varying R and K ratios. Notably, the thermal conductivity decreased with higher temperature. According to Kingery [38,39], the phonon vibration is the main heat transfer mechanism in a ceramic system, and such a heat transfer through phonon vibration can be expressed by the following equation

$$\lambda = \frac{1}{3}Cvl \tag{1}$$

where C is the specific heat capacity, v is the mean particle velocity, and l is the phonon mean free path of collision. Above the Debye temperature (100–1000 K) [39], the specific heat capacity (C) and mean particle velocity (v) become approximately constant, [38–40] and the thermal conductivity of the molten oxide system is mainly affected by the change of the mean free path of collision (l), which is proportional to the inverse of the temperature. Therefore, the observed negative temperature dependence of the thermal conductivity would result from the decrease in the phonon mean free path of collision (l), as described by Eq. (1).

3.2. Effect of the R and K ratios on the thermal conductivity

The change of thermal conductivity with varying Na_2O/B_2O_3 ratio at 1273 K is shown in Fig. 3. The conflicting effect of the R ratio on thermal conductivity can be observed at the fixed values of K = 0.5 and 1.0. Thermal conductivity initially increases with increasing R ratio, but after reaching a maximum at approximately R = 0.5, it decreases. On the other hand, at K = 2.0, only a monotonous decrease in thermal conductivity can be observed in the current work. Such a change of thermal conductivity with varying R ratio is in good agreement with the change of 4-coordinated boron. In addition, a higher thermal conductivity can be observed at higher K ratios across the entire range of R ratios, indicating a significant effect of SiO_2 on thermal conductivity.

3.3. Structural investigation of the sodium borosilicate system using ^{11}B and ^{29}Si MAS NMR

In order to find out the relationship between the relative fraction of 4-coordinated boron and thermal conductivity, borate structure was investigated using ^{11}B MAS NMR. Compared to other infrared spectroscopy, ^{11}B MAS NMR has the virtue of readily distinguishing between 3- and 4-coordinated boron. In addition, the silicate network structure was evaluated using ^{29}Si MAS NMR. Due to the weak Raman intensity of Q^4 unit in the high frequency region [41], Raman spectroscopy is not suitable for the present $Na_2O–B_2O_3–SiO_2$ system consisting of a large proportion of Q^4 unit.

Fig. 4 shows the ^{11}B MAS NMR spectra of the current $Na_2O–B_2O_3–SiO_2$ system. Du and Stebbins [3] studied the structure of the sodium borosilicate system reporting that each 3-coordinated boron and 4-coordinated boron is centered on 12 and 0 ppm, respectively. Nanba et al. [42] found that the asymmetric broad peak observed between 20 and −20 ppm can be attributed to 3-coordinated boron in the BO_3 unit. On the basis of the references [3,42], the present MAS NMR spectra were identified and the relative fractions of 3- and 4-coordinated boron were calculated based on the area ratio.

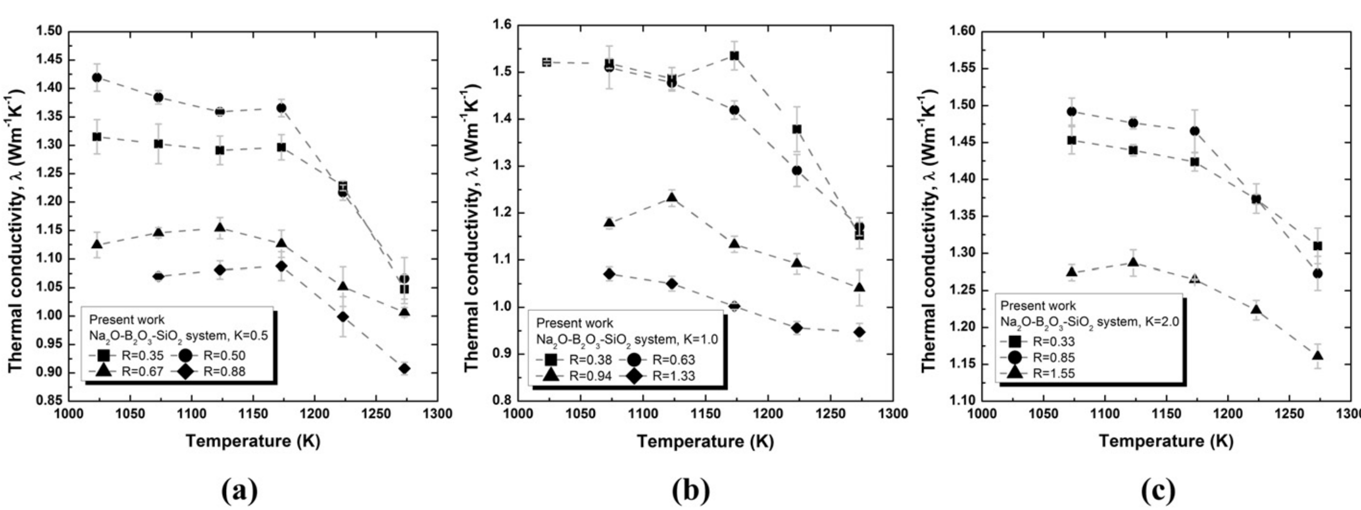

Fig. 2. Temperature dependence of the thermal conductivity in the $Na_2O–B_2O_3–SiO_2$ system for (a) K = 0.5, (b) K = 1.0, and (c) K = 2.0. (The gray dashed line has been drawn as a guide).

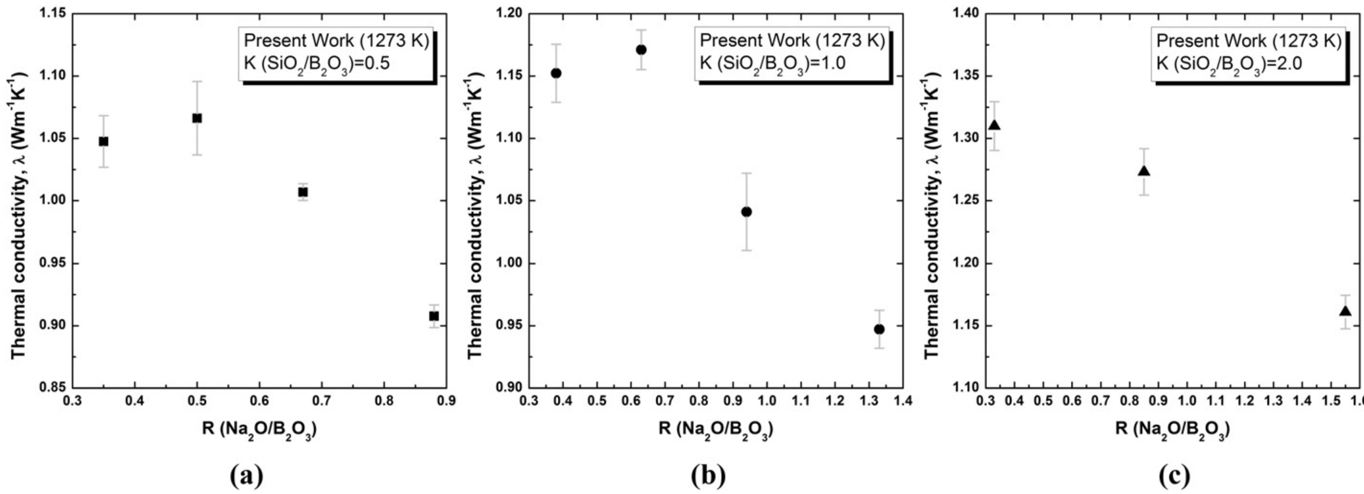

Fig. 3. Relationship between the R ratio and the thermal conductivity of the Na_2O–B_2O_3–SiO_2 system at 1273 K for different K ratios: (a) K = 0.5, (b) K = 1.0, and (c) K = 2.0.

Fig. 5 shows the change of 4-coordinated boron with varying R ratio at a fixed K ratio. Although the relative fraction of 4-coordinated boron obtained in the present work is about 5 pct point higher than that in the previous work [42], the overall tendency shows an excellent accordance between the values. The relative fraction of 4-coordinated boron initially increases with increasing R ratio, reaches a maximum around R = 0.6–0.8, and then decreases.

Fig. 6 shows the ^{29}Si MAS NMR spectra of the present Na_2O–B_2O_3–SiO_2 system with varying R and K ratios. According to Maekawa et al. [43], who studied the alkali-silicate glass structure using ^{29}Si MAS NMR, the Q^4, Q^3, and Q^2 units of the silicate network are observed within the ranges −110 through −95 ppm, −92 through −85 ppm, and −80 through −75 ppm, respectively. Similar chemical shifts of the Q^n species for ^{29}Si MAS NMR were reported by Nanba et al. [42], who measured and calculated the sodium borosilicate glasses structure using MAS NMR and first-principle molecular orbital (MO) calculation. Based on the references [42,43], the present ^{29}Si MAS NMR spectra were identified and the Gaussian deconvolution was carried out.

The change in the relative fraction of each Q^n species for the silicate structure with varying R and K ratios is shown in Fig. 7. Compared to the previous work [42], a similar structure change of the silicate network can be observed. The increase in the Na_2O/B_2O_3 ratio results in the depolymerization of the silicate network.

Owing to the considerable change of the boron coordination number at different temperatures in the molten state, many researchers have focused on the relationship between the borate structure change and the quenching rate along with the fictive temperature (T_f), where the glass structure is in equilibrium [5,44–46]. In addition, Yano et al. [47] reported the change of the silicate structure at different temperatures in the molten alkali silicate system using high-temperature Raman spectroscopy. It should be noted that the present structural investigations carried out in the quenched glass system could not provide exact structural information at 1273 K. However, assuming that the fictive temperature (T_f) and the structural change rate are similar across the compositional range, it is worth considering the present ^{11}B and ^{29}Si MAS NMR results to qualitatively evaluate the effect of the structure on thermal conductivity.

4. Discussion

4.1. Consideration of one-dimensional Debye temperature (Θ_{D1})

In the low-temperature region, the thermal conductivity of the oxide system initially increases with increasing temperature as a result of the increase in both the specific heat capacity (C) and the mean particle velocity (v), as described by Eq. (1) [32,48]. However, above the

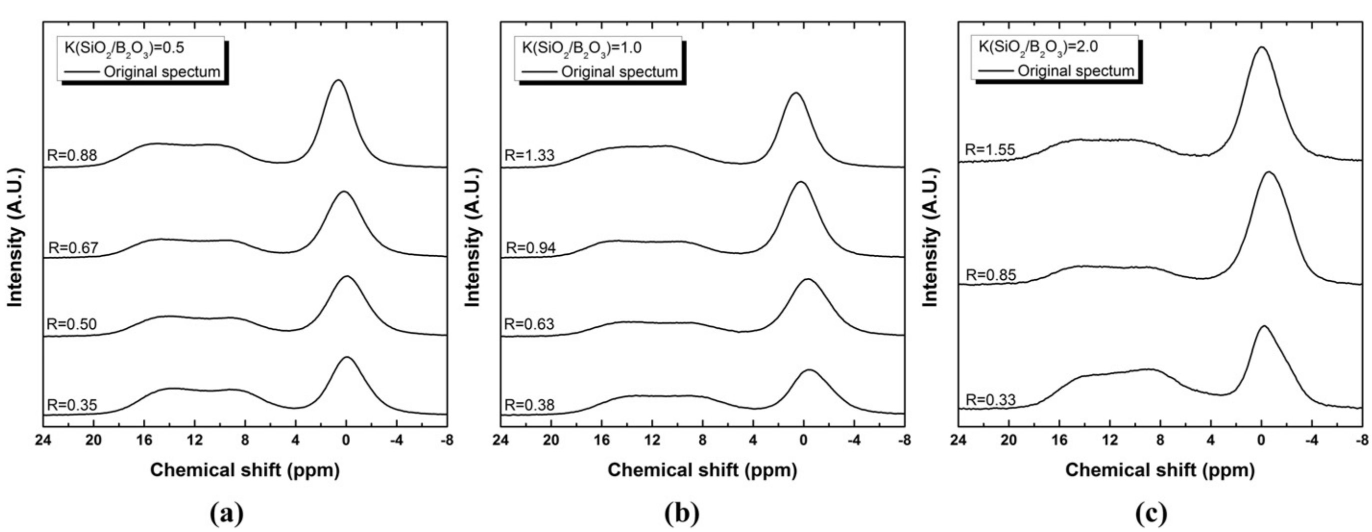

Fig. 4. ^{11}B MAS NMR spectra with varying R ratio for (a) K = 0.5, (b) K = 1.0, and (c) K = 2.0.

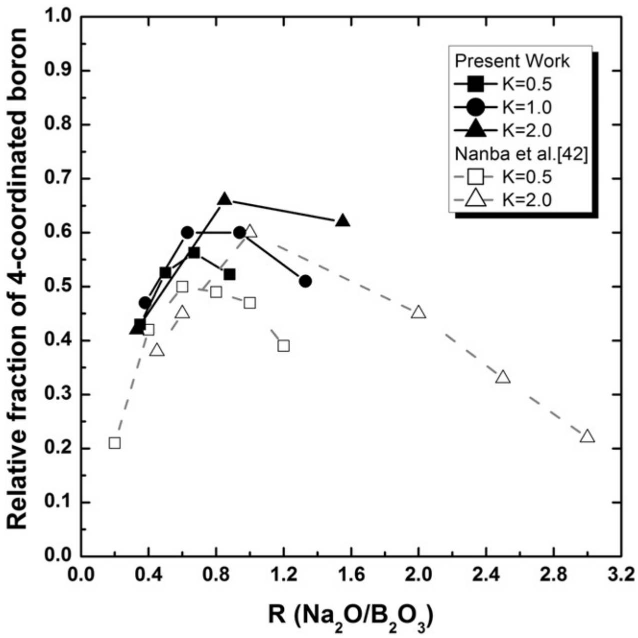

Fig. 5. Relationship between the R ratio and the relative fraction of 4-coordinated boron at a fixed K ratio. (The black solid and gray dashed lines have been drawn as a guide.)

Debye temperature, owing to the approximately constant specific heat capacity (C) and mean particle velocity (v), the thermal conductivity of the oxide system decreases with increasing temperature as a result of the decrease in the phonon mean free path of collision (l), which is proportional to the inverse of the temperature ($1/T$). [25–29,31] Hence, it can be inferred that thermal conductivity reaches the maximum value around the Debye temperature, and then it decreases with higher temperatures. Therefore, identifying the Debye temperature in the oxide system is essential to evaluate the temperature at which thermal conductivity shows the maximum and to determine which variables mainly affect the thermal conductivity at the given temperature.

According to Inaba et al. [49], the real macromolecular structure of a highly polymerized glass system requires not only a three-dimensional continuum as postulated by the Debye theory, but also a one-dimensional continuum. They found that the heat capacity of the oxide systems follows the one-dimensional Debye model when the temperature ranges between 300 K and the glass transition temperature. Based on the ionic packing ratio (V_P) and the dissociation energy per unit volume (G_t), they proposed an equation to calculate the one-dimensional Debye temperature [49]

$$\Theta_{D1} = \left(\frac{3}{2}\right)\left(\frac{h}{k}\right)\left(\frac{N_A}{V_a}\right)^{1/3}\sqrt{\frac{2V_P G_t}{\rho}} \tag{2}$$

where Θ_{D1} is the one-dimensional Debye temperature (K), h is the Planck constant (6.63×10^{-34} $m^2 \cdot kg \cdot s^{-1}$), k is the Boltzmann constant (1.38×10^{-23} $m^2 \cdot kg \cdot s^{-2} \cdot K^{-1}$), N_A is the Avogadro's number (6.02×10^{23} mol^{-1}), V_p is the ionic packing ratio, G_t is the dissociation energy (J/m^3), and ρ is the density (kg/m^3).

Using Eq. (2), the one-dimensional Debye temperature of the present system was calculated. In this calculation, the density was estimated from reference [19]. In addition, the relative fraction of both 3- and 4-coordinated boron was also considered. Fig. 8 shows the one-dimensional Debye temperature of the current Na_2O–B_2O_3–SiO_2 system with varying R and K ratios. The one-dimensional Debye temperature of the present system can be found between 920 K and 1060 K. In this work, all the measurements were carried out above the one-dimensional Debye temperature, indicating that specific heat capacity and mean particle velocity could be characterized as constant. Therefore, the thermal conductivity of the present system is mainly affected by the change of the mean free path of collision (l), which is inversely proportional to the temperature, resulting in the decrease in thermal conductivity with increasing temperature.

In Fig. 8, the one-dimensional Debye temperature decreases with higher R or lower K ratio. According to Inaba et al. [49], the one-dimensional Debye temperature is related to the interatomic bonding between network forming cations and anions. Therefore, the decrease in the one-dimensional Debye temperature with higher R or lower K ratios reflects the weakening of the bonding strength. Because of the formation of 4-coordinated boron, which has a weaker bonding strength than 3-coordinated boron [50], and the depolymerization of the silicate network, the bonding strength decreases at higher R ratios reducing the one-dimensional Debye temperature.

4.2. Effect of the borate and silicate structure on thermal conductivity

As shown in Fig. 5, 4-coordinated boron increases until it reaches an R ratio of 0.6–0.8. According to the previous Raman spectroscopy studies concerning the glass Na_2O–B_2O_3–SiO_2 system [51], the increase in 4-coordinated boron is related to the formation of diborate and tetraborate units, which have a 3-dimensional network structure. Although the silicate network structure is gradually depolymerized at higher R ratios, it

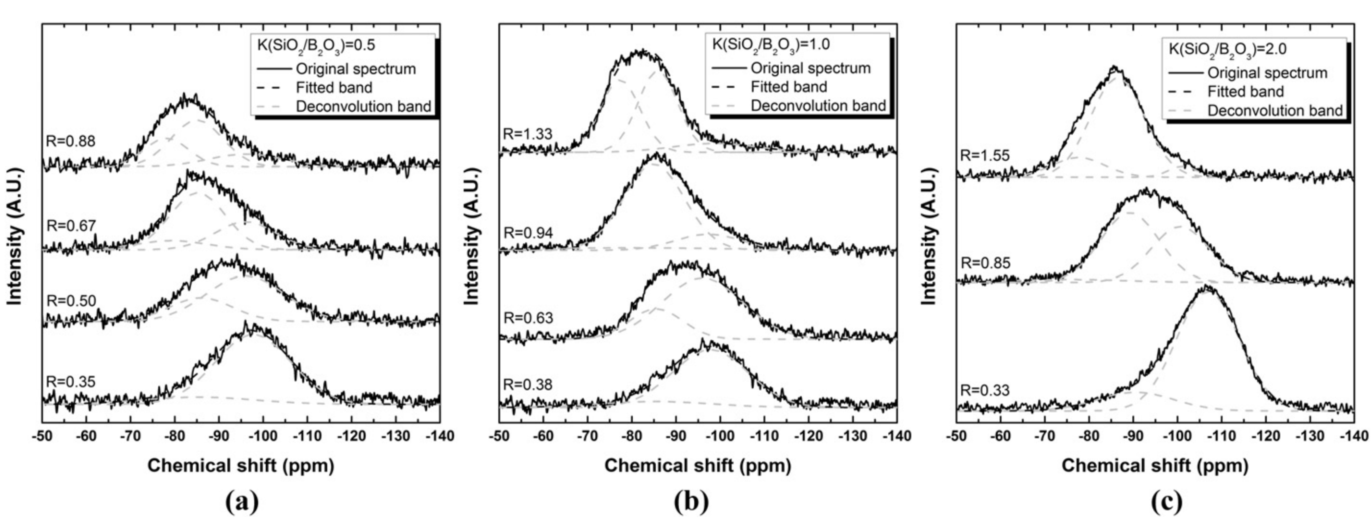

Fig. 6. ^{29}Si MAS NMR spectra with varying the R ratio for (a) K = 0.5, (b) K = 1.0, and (c) K = 2.0.

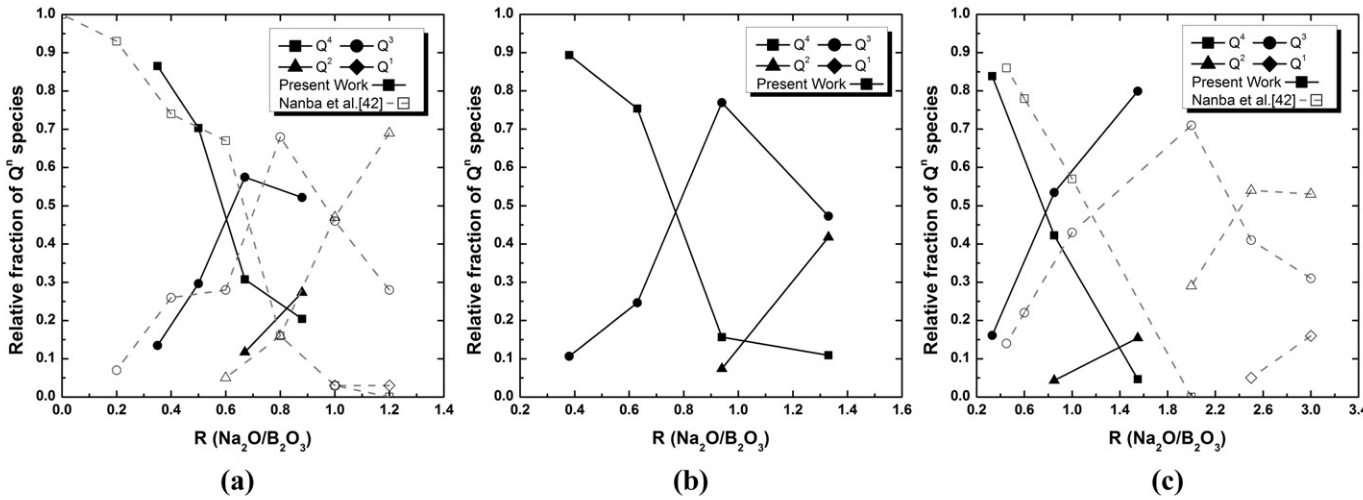

Fig. 7. Relationship between the R ratio and the relative fraction of the Q^n species for (a) K = 0.5, (b) K = 1.0, and (c) K = 2.0. (The black solid and gray dashed lines have been drawn as a guide.)

seems that the extent of the Q^n species change is relatively small; less than 15 pct point, in the region where R is lower than 0.5 for K = 0.5, and than 0.6 for K = 1.0. Recently, Inoue et al. [13] calculated the structure of the glass Na_2O–B_2O_3–SiO_2 system using pair potentials by means of the MD technique. Within the R ratio range of 0–0.5 at the fixed values of K = 0.5 and 2.0, they found that the relative fraction of NBO in the Si–O and B–O bonds is kept almost constant, being less than 0.05. Therefore, below R = 0.5, the depolymerization of the silicate network structure is relatively insignificant, but the change of the borate structure from 3- to 4-coordinated boron is dominant. Considering the formation of the 3-dimensional network structure along with the increase in 4-coordinated boron, the structure of the molten Na_2O–B_2O_3–SiO_2 system becomes more polymerized at higher R ratios, resulting in an increase in the phonon mean free path of collision (l). For this reason, the thermal conductivity of the present system initially increases with increasing R ratio (Fig. 3).

On the other hand, as shown in Fig. 3, the thermal conductivity of the system gradually decreases above the R ratio of 0.5–0.6. Although the relative fraction of 4-coordinated boron increases within the R ratio range of 0.6–0.8, it seems that thermal conductivity is not significantly affected by the change of 4-coordinated boron. As shown in Fig. 7, a drastic decrease in the Q^4 species along with the increase in the Q^3 species can be observed above the R ratio of 0.5–0.6, indicating the significant depolymerization of the silicate network. According to Inoue et al. [13], a drastic increasing of the fraction of the NBOs in the B–O and Si–O networks can be observed above R = 0.5 at the fixed K = 0.5 and 2.0. Therefore, owing to the formation of NBO and the depolymerization of the silicate structure, thermal conductivity gradually decreases when the R ratio increases above 0.5–0.6.

Fig. 9 shows the relationship between the change of the K ratio and thermal conductivity obtained for the R ratio in the range of 0.33–0.38 at 1273 K. Thermal conductivity gradually increases with increasing the K ratio. Considering the similar relative fraction of 4-coordinated boron and the similar relative fraction of Q^n species; 0.84–0.89 for the Q^4 unit and 0.11–0.16 for the Q^3 unit, it seems that the change in thermal conductivity with varying the K ratio results from the substitution of SiO_2 for B_2O_3. A similar result, namely, an increase in thermal conductivity at higher SiO_2 content, has been reported by Ghoneim and Halawa [18], who measured thermal conductivity in the glass Na_2O–B_2O_3–SiO_2 system. They inferred that the formation of a rigid and compact structure by increasing the SiO_2 content results in a longer phonon mean free path of collision (l) along with an increase in thermal conductivity.

According to Du and Stebbins [3], who studied the glass Na_2O–B_2O_3–SiO_2 structure using ^{11}B and ^{17}O NMR, the average mean number of Si coordinated to 4-coordinated boron increases for higher Si/(Si + B) ratios. Recently, Koroleva and Shabunina [52] investigated glass structure in the Na_2O–B_2O_3–SiO_2 system using Raman spectroscopy. According to their simulation of high-frequency Raman spectrum, Q^{34} silicate structure; Q^3 unit associated with Q^4 unit, increases with higher K ratio at a fixed R ratio. In addition, both ^{17}O triple quantum (3Q) MAS [3] and MD calculation [13] observed the increase in the Si–O–Si bond at higher K ratios in the Na_2O–B_2O_3–SiO_2 system. Therefore, thermal conductivity increases for higher K ratios because of the increase in the total amount of the silicate network structure and the more effective mixing of Si and 4-coordinated boron.

5. Conclusions

Using the hot-wire method, thermal conductivity of the molten Na_2O–B_2O_3–SiO_2 system was measured across a temperature range from 1273 K to its liquidus temperature with varying R ratio (Na_2O/B_2O_3) at a fixed K ratio (SiO_2/B_2O_3). Thermal conductivity decreased at higher temperature because of the shortening of phonon mean free path of collision (l). The one-dimensional Debye temperature (Θ_{D1}) of the present sodium

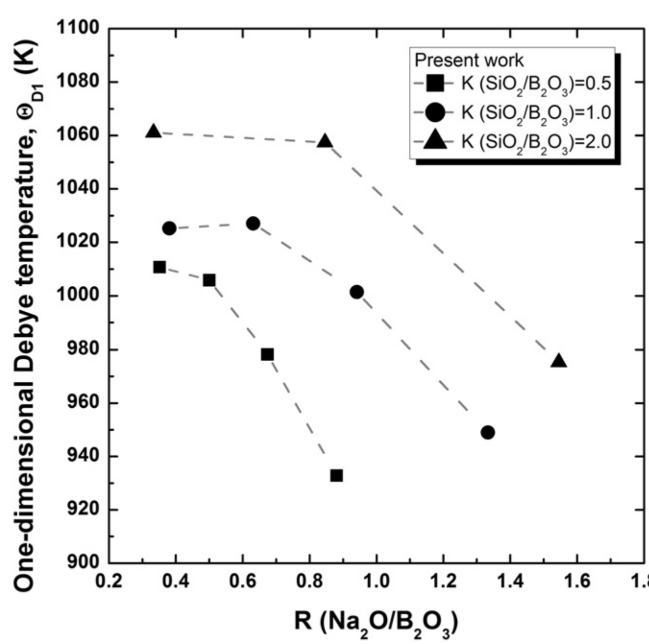

Fig. 8. Relationship between the R ratio and the calculated one-dimensional Debye temperature of the present system at a fixed K ratio. (The gray dashed line has been drawn as a guide.)

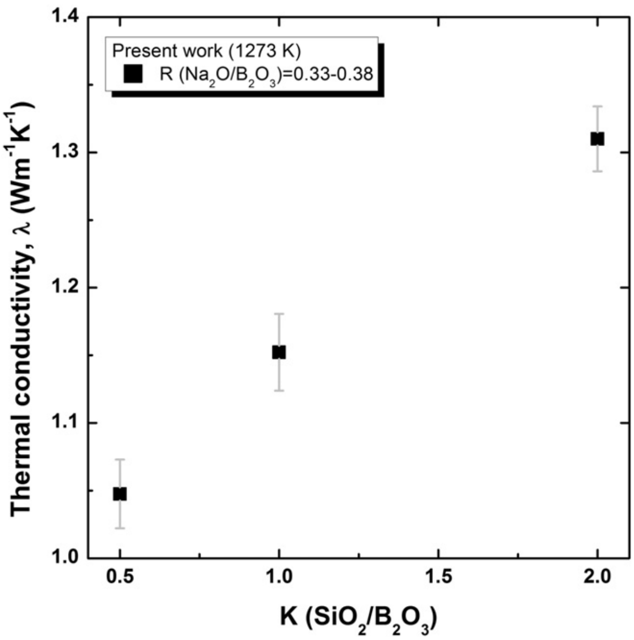

Fig. 9. Relationship between the K ratio and the thermal conductivity of the present system at 1273 K.

borosilicate system was calculated. The one-dimensional Debye temperature decreases at higher R or lower K ratio, reflecting the change in the bonding strength along with the composition change. Using the ^{11}B and ^{29}Si MAS NMR, the structure of the glass Na_2O–B_2O_3–SiO_2 system was investigated. It was observed that the relative fraction of 4-coordinated boron initially increases with increasing R ratio, and then subsequently decreases. On the other hand, the silicate network structure was gradually depolymerized with increasing R ratio. At 1273 K, the conflicting effect of the R ratio on thermal conductivity was observed for the fixed K ratios of 0.5 and 1.0. Thermal conductivity initially increased with increasing R ratio, but gradually decreased above R = 0.5. Thus, owing to the increase in 4-coordinated boron along with the formation of three-dimensional network units, thermal conductivity increases. However, for an R ratio above 0.5, owing to the drastic increase in NBO along with depolymerization of silicate network, thermal conductivity decreases with increasing R ratio. Finally, the effect of the K ratio on thermal conductivity was evaluated. Owing to the increase in the total amount of the silicate network structure and the better mixing between 4-coordinated boron and silicon, thermal conductivity increases at higher K ratios.

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